

Daniel Lopez

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EDUCATION

Stony Brook University

Stony Brook, NY

Bachelor of Science in Data Science and Computer Science Program

Jan. 2026 – May 2028

- Relevant Coursework: Data Structures, Object Oriented Programming, Data Science, Discrete Math, Visualization

TECHNICAL SKILLS

Programming: Java, Python, SQL, TypeScript

Frameworks & Tools: React, Next.js, Node.js, Electron, Three.js, Postgres, SQLite, Drizzle, Anthropic SDK, MCP, Ollama, Git, Claude Code, Adobe Creative Suite

EXPERIENCE

Research Assistant, Frontend Developer

Aug. 2026 – Present

PoliTech, Stony Brook University (Supervisor: Prof. Robert Kelly)

Stony Brook, NY

- Built a seat-share curve tool comparing single- and multi-member districting plans for the Fair Representation Act.
- Visualized partisan fairness across generated plan ensembles, contrasting single-member districts with ranked-choice voting.
- Connected the GUI to the ranked-choice voting simulator and district ensembles by working with the data team.

IT Systems Technician

Feb. 2026 – May 2026

Stony Brook University Department of Information Technology

Stony Brook, NY

- Resolved 50+ faculty and student tickets per semester in TeamDynamix by troubleshooting software and network issues.
- Reimaged and deployed 200+ computers using Jamf and Ghost, standardizing software and security configurations.
- Maintained printer infrastructure across the University, reducing downtime through maintenance and network management.

Video & Graphic Design Editor

Aug. 2023 – Dec. 2024

The College Soccer Guy (Remote)

Los Angeles, CA

- Delivered polished video content for 20+ athlete clients on-deadline using Adobe Premiere Pro and Photoshop.
- Grew the agency's Instagram to 115K followers by designing graphics for posts.

PROJECTS

FutTraffic — Computer Vision Football Analytics | *Python, PyTorch, OpenCV, FastAPI, Next.js, Electron, Anthropic SDK*

- Built a desktop app turning one fixed match camera into per-player distance, speed, and sprint data, in partnership with an MLS NEXT academy.
- Developing an RF-DETR pipeline turning positional data into per-player and team events, toward parity with manual tagging.
- Built an LLM agent-driven coaching workspace with automated player and team clipping, a report notepad, and a player database.

DreamByte — Agentic Video Editor 🧙 | *TypeScript, React, Electron, Three.js, MCP, FFmpeg*

- Built an Electron NLE where a director agent with 83 tools turns prompts into multi-scene animated videos across 10 renderers, with every layer editable on a timeline.
- Wrote a provider-agnostic LLM layer spanning 9 backends (Anthropic, OpenAI, Google, DeepSeek, Qwen, Kimi, Ollama, Claude Code, Codex CLI) and a write-ahead action log driving undo/redo and Git-backed scene branching.
- Unified 30+ generative-media providers (image, video, avatar, TTS, music, SFX) behind one API; shipped FFmpeg MP4 export, interactive publishing, and an MCP server exposing the editor to Claude Code.

Llamas Remote — Local-First macOS AI Notebook 🧙 | *TypeScript, Electron, React, TipTap, SQLite FTS5, Ollama, Swift*

- Built a macOS notch HUD that captures selections or screen regions and streams Ollama, OpenAI, or Anthropic answers into notes.
- Built a Markdown-as-truth notebook with SQLite FTS5 search, local RAG chat over notes, a calendar, and a photo library.
- Engineered the Electron and React app on TipTap, Excalidraw, and SQLite, with a multi-provider LLM router and on-device Swift Vision OCR.

Agentic Point-of-Sale 🧙 | *Python, Flask, React, PostgreSQL, Tauri, Claude API, Stripe Terminal, Socket.IO*

- Built a full-stack, multi-location POS platform replacing separate checkout, inventory, accounting, and scheduling tools with one system.
- Eliminated manual data entry by extracting line items from vendor invoices (PDF, image, spreadsheet) via Claude and GPT-4o vision, auto-restocking inventory and syncing to Shopify.
- Built the React POS client, Flask and Socket.IO APIs, Tauri desktop app, and Stripe Terminal card checkout, with Daniel Budnyatsky co-building the accounting and scheduling backend.

HONORS & AWARDS

2024 – NYC Public School Athletic League All City Team (22 awardees out of ~9,000 athletes)

2023 – Certificate of Recognition, New York City Comptroller